



Sec: Sr. IIT

Race -12

Date: 27-02-24

PHYSICS
SECTION - I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct**.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

1. The viscosity of a fluid μ can be determined by measuring the terminal velocity V_T of a sphere when it descends in the fluid. The fluid has a density ρ_f while the sphere has a density ρ_s and a diameter of d . The viscosity can then be calculated by the formula

$$\mu = \frac{5(\rho_s - \rho_f)}{9V_T} d^2$$

The values measured are

$$V_T = (1.60 \pm 0.04) \text{ ms}^{-1}$$

$$\rho_s = (2700 \pm 20) \text{ kg m}^{-3}$$

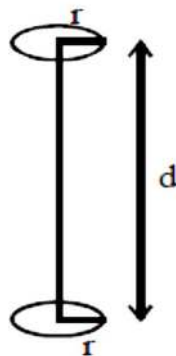
$$\rho_f = (900 \pm 10) \text{ kg m}^{-3}$$

$$d = (20.0 \pm 0.4) \text{ mm}$$

What is the approximate percentage uncertainty in the value of μ ?

- A) 6.2% B) 7.1% C) 8.2% D) 9.3%

2. Two thin identical current carrying rings having radius ' r ' and current i in each are separated by distance ' d ' as shown in the diagram. If $r \ll d$, then find the magnetic force between the two rings.



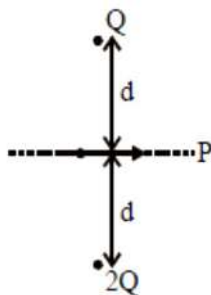
- A) $\frac{6\mu_0}{4\pi} \frac{i^2 \pi^2 r^4}{d^4}$ B) $\frac{\mu_0}{4\pi} \frac{i^2 \pi^2 r^4}{d^4}$ C) $\frac{3\mu_0}{4\pi} \frac{i^2 \pi^2 r^4}{d^4}$ D) $\frac{\mu_0}{2\pi} \frac{i^2 \pi^2 r^4}{d^4}$

3. Superconductors have a property that magnetic flux linked with them will always remain constant. Two identical superconducting rings each having self inductance " L " carrying current ' I ' in same direction are placed together as shown in the diagram. Find work done against magnetic interaction in separating the rings to far away [infinite distance]. (Assume that the mutual inductance of the two rings is $M = L$ in the initial position).



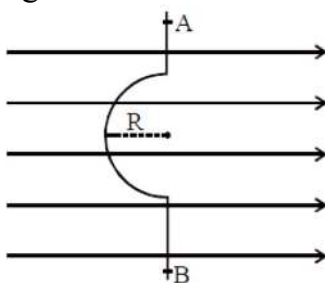
- A) LI^2 B) $\frac{1}{2}LI^2$ C) $2LI^2$ D) $-LI^2$

4. Two point charges 'Q' and '2Q' are placed at a distance 'd' on equatorial line of dipole of dipole moment 'p' as shown in diagram. Find net force acting on the dipole due to point charges. (where $d \gg$ size of dipole)



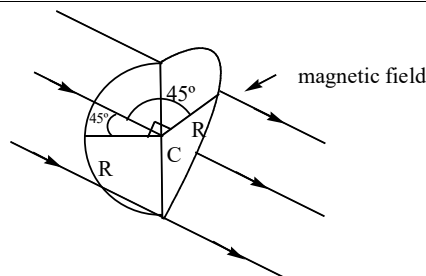
- A) 0 B) $\frac{kpQ}{d^3}$ C) $\frac{3kpQ}{d^3}$ D) $\sqrt{5} \frac{kpQ}{d^3}$

5. Figure shows a rigid thin frame, hinged at A and B with the help of two mass less and charge less rods and given the shape of half ring of radius R. The assembly is now free to rotate about vertical axis passing through AB in a horizontal uniform electric field E. Linear charge density of frame is λ_1 and linear mass density is λ_2 . If assembly is slightly disturbed from unstable equilibrium position, then find maximum angular velocity of ring during ensuing motion.



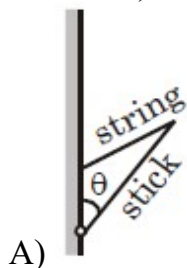
- A) $\sqrt{\frac{16\lambda_1 E}{\pi\lambda_2 R}}$ B) $\sqrt{\frac{\pi}{16} \frac{\lambda_1 E}{\lambda_2 R}}$ C) $\sqrt{\frac{\lambda_1 E}{2\lambda_2 R}}$ D) $\sqrt{\frac{2\lambda_1 E}{\lambda_2 R}}$

6. Figure shows a closed loop of wire that consists of a pair of equal semicircles of radius R, lying in mutually perpendicular planes. The loop was formed by folding a flat circular loop along a diameter until the two halves became perpendicular to each other. A uniform magnetic field of magnitude B is directed perpendicular to the fold diameter and makes equal angles (of 45°) with the planes of the semicircles, initially ($t = 0$). The loop is rotated at a constant angular velocity ω about the fold diameter. This induces an emf, which causes a current to flow in the wire. The resistance per unit length of the wire is λ . The magnetic moment of the loop is

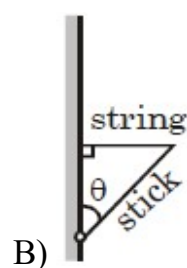


A) $\left(\frac{B\pi R^3 \omega}{2\sqrt{2}\lambda} \sin \omega t \right)$ B) $\left(\frac{B\pi R^3 \omega}{2\lambda} \cos \omega t \right)$ C) $\left(\frac{B\pi R^3 \omega}{4\lambda} \cos \omega t \right)$ D) $\left(\frac{B\pi R^3 \omega}{4\lambda} \sin \omega t \right)$

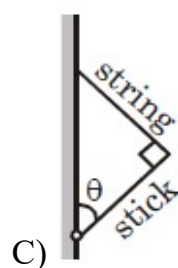
7. A stick of uniform mass distribution has its bottom end attached to a wall by a pivot and is held up by a massless string attached to its other end. Which of the following scenarios has the smallest tension in the string? (Length of stick is same in all scenarios).



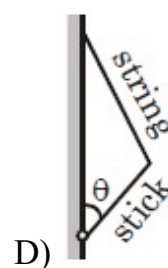
A)



B)



C)



D)

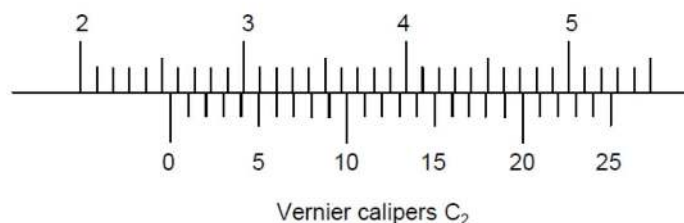
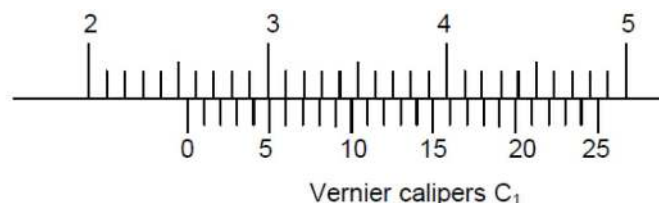
SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

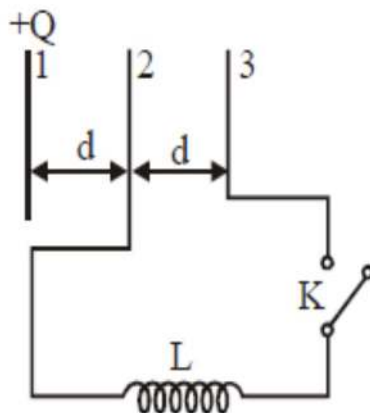
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

8. There are two Vernier Calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers has 25 equal divisions coinciding with 27 divisions on the main scale; while the Vernier scale of the other has 25 divisions coinciding with 23 divisions on the main scale. You have to decide which pair of data correspond to which calipers by referring to the figures below:

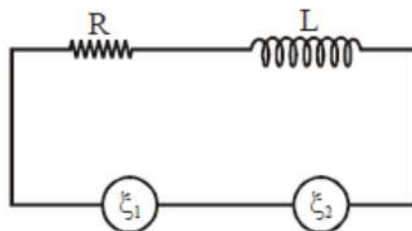


- A) The smallest reading that can be taken on C_1 is 0.004 cm.
 B) The smallest reading that can be taken on C_2 is 0.008 cm.
 C) The reading indicated by C_2 is 2.552 cm.
 D) The reading indicated by C_1 is 2.548 cm.

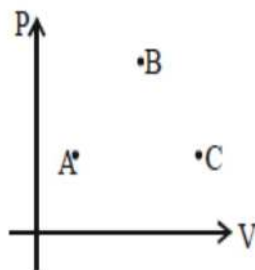
9. Three identical thin, fixed metal plates are arranged in the air at equal distances d from each other. The area of each of the plates equals A . The plate 1 has charge $+Q$, while plates 2 and 3 are not charged and are connected through a key K to an ideal inductor of self inductance L . The distance between the plates is small compared with their size. Select correct option(s)



- A) Just before the switch K is closed, the charge on right surface of plate 2 is $Q/2$
 B) Just before the switch K is closed, the charge on right surface of plate 3 is $Q/2$
 C) After the switch K is closed, the maximum current in the circuit is $\frac{Q}{2} \sqrt{\frac{d}{L\epsilon_0 A}}$
 D) After the switch K is closed, the maximum current in the circuit is $Q \sqrt{\frac{d}{L\epsilon_0 A}}$
10. Two AC sources with emfs $\xi_1 = 10 \sin(\omega t) \text{ V}$ and $\xi_2 = 10 \sin\left(\omega t - \frac{\pi}{3}\right) \text{ V}$ are fed to an LR circuit. The resistance is $R = 10 \Omega$ and inductive reactance is $X_L = 10\sqrt{3} \Omega$

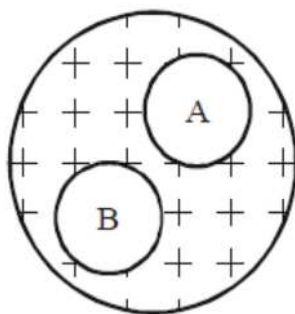


- A) The equivalent emf of the AC source has a peak value of $10\sqrt{3} \text{ V}$
 B) The peak value of current is $\frac{\sqrt{3}}{2} \text{ A}$
 C) The voltage across the inductor at $t = \frac{\pi}{3\omega}$ is $\frac{15\sqrt{3}}{2} \text{ V}$
 D) The voltage across the resistor at $t = \frac{\pi}{3\omega}$ is $\frac{15}{2} \text{ V}$
11. An ideal gas can expand from state A to C (via state B) through two different process P and Q. In process P, AB and BC are straight lines in P-V diagram. In process Q, states A, B, C are on a circle in P-V diagram. Heat supplied and work done by gas in processes P and Q are respectively Q_P, Q_Q and W_P, W_Q

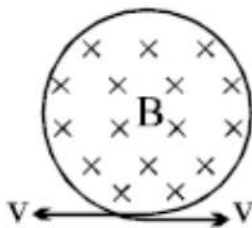


- A) $Q_Q > Q_P$ B) $W_Q < W_P$ C) $\frac{W_Q}{Q_Q} > \frac{W_P}{Q_P}$ D) $Q_Q - W_Q = Q_P - W_P$

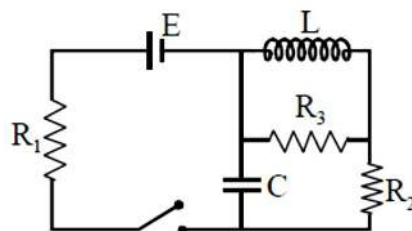
12. A solid non conducting sphere have two spherical cavities A and B of radius ' r_A ' and ' r_B ' respectively, as shown. Radius of sphere is ' R '. Select the **CORRECT** statement(s):-



- A) Electric field in cavity A is uniform.
 B) Electric field at the centre of cavity A depends on radius of cavity A.
 C) Electric field at the centre of cavity A depends on radius of cavity B.
 D) Electric field at the centre of cavity A will depend on distance between centres of cavities A and B.
13. A circular conducting loop of radius r_0 and having resistance per unit length λ as shown in the figure is placed in a magnetic field B which is constant in space and time. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing, then



- A) radius of the loop changes with r as $r = r_0 - vt / \pi$
 B) magnitude of EMF induced in the loop as a function of time is $e = 2Bv|r_0 - vt / \pi|$
 C) current induced in the loop is $I = \frac{Bv}{2\pi\lambda}$
 D) current induced in the loop is $I = \frac{Bv}{\pi\lambda}$
14. Current in R_3 in the shown circuit (capacitor is initially uncharged)



- A) just after closing the switch is zero
 B) long time after closing the switch is zero
 C) just after closing the switch is $\frac{E}{R_3}$
 D) long after closing the switch is $\frac{E}{R_3}$

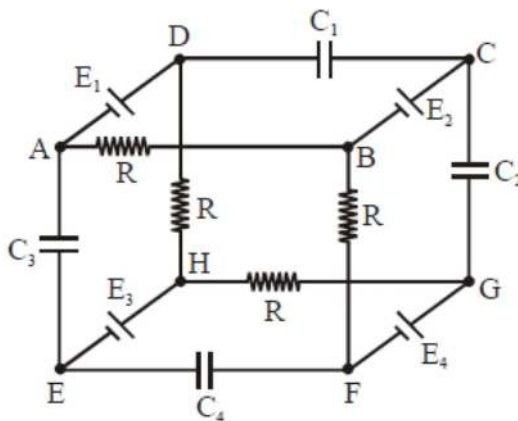
SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16

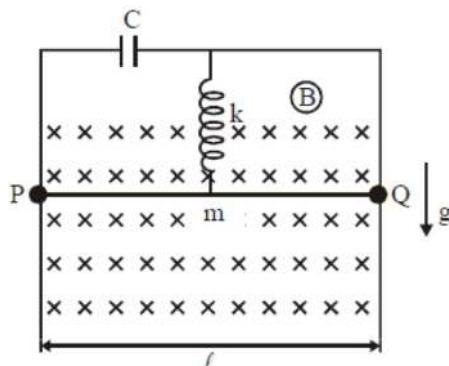
Four batteries of emf $E_1 = 4V$, $E_2 = 8V$, $E_3 = 12V$ and $E_4 = 16V$ and four identical capacitors each of $1\mu F$ and four identical resistances are connected in circuit as in figure. Batteries do not have any internal resistances.



15. At steady state, potential difference across C_2 is:
 A) 1V B) 5V C) 4V D) 6V
16. If points H and B are shorted, the new charge on C_2 at steady state will be:
 A) Zero B) $5\mu C$ C) $4\mu C$ D) $1\mu C$

Paragraph For Questions 17 and 18

A conducting (resistance less) slider PQ of mass m and length ℓ is free to translate on two vertical smooth rails in the presence of gravity. The slider is connected with a non conducting spring of spring constant k in uniform magnetic field B into the plane. At time $t = 0$, spring was in its natural length, capacitor was uncharged and the slider is released from rest as shown in the figure, then answer the following questions.



17. Motion of rod will be:-

- A) SHM with angular frequency $\sqrt{\frac{k}{m}}$
 B) oscillatory but not SHM
 C) SHM with angular frequency $\sqrt{\frac{k}{m + B^2 \ell^2 C}}$
 D) SHM with amplitude of oscillation of rod as $\frac{4mg}{k}$

18. If suddenly spring is cut while rod is in motion, then:

- A) the rod will still perform oscillatory motion
 B) the rod will move downward and acquire terminal velocity
 C) finally, the rod moves down with constant acceleration
 D) charge on capacitor will increase with time

CHEMISTRY

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct**.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

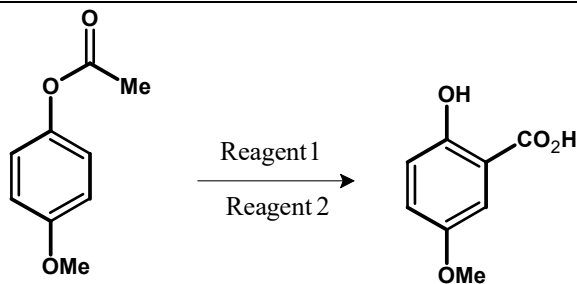
19. From the relations between the variables for two gases, given in List - II and the conclusion regarding the variables on the List I

List I		List-II	
(P)	$\lambda_1 > \lambda_2$	(1)	Equal P, $T_1 = 2T_2, \sigma_2 = 2\sigma_1$
(Q)	$Z_1(1) > Z_1(2)$	(2)	Equal T and P, $M_1 = 2M_2, \sigma_1 = 2\sigma_2$
(R)	$Z_{11}(1) < Z_{11}(2)$	(3)	Equal T and P, $M_1 = 2M_2, \sigma_2 = 2\sigma_1$
(S)	$K.E_1 = K.E_2$	(4)	Equal T, P and V, $M_1 > M_2$

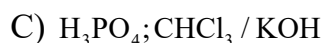
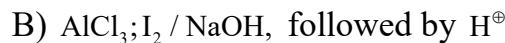
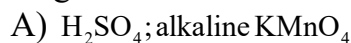
Match List I with one or more options of List II. The correct option is

- A) $P \rightarrow 1$ $Q \rightarrow 2$ $R \rightarrow 3$ $S \rightarrow 4$
 B) $P \rightarrow 1$ $Q \rightarrow 3$ $R \rightarrow 2$ $S \rightarrow 4$
 C) $P \rightarrow 4$ $Q \rightarrow 2$ $R \rightarrow 3$ $S \rightarrow 1$
 D) $P \rightarrow 4$ $Q \rightarrow 3$ $R \rightarrow 2$ $S \rightarrow 1$

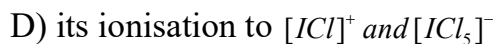
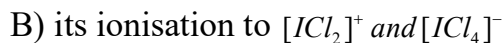
20. In the following transformation



reagents 1 and 2 are:



21. ICl_3 is unstable and it dimerises to I_2Cl_6 . The liquid has an appreciable electrical conductivity while in gas phase it decomposes into ICl and Cl_2 . The reason for the observed electrical conductivity is:



22. Light of frequency ν is found to eject electrons of velocity v_e from a clean potassium surface in vacuum. Which of the following is true concerning this phenomenon?

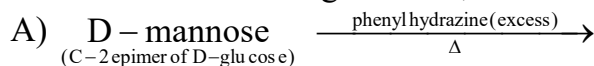
A) This phenomenon is best explained theoretically by using the wave model of light.

B) Light of frequency 2ν will eject electrons of velocity $2v_e$.

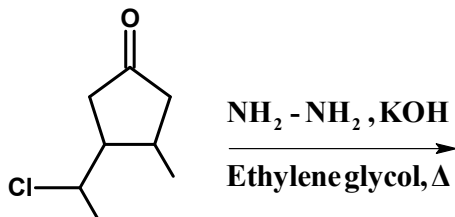
C) A more intense light source of frequency ν will eject electrons with a velocity greater than v_e .

D) The minimum energy required to remove an electron from the metal is $h\nu - \frac{1}{2}mv_e^2$.

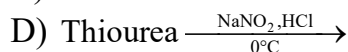
23. Which of the following reaction, do not evolve nitrogen gas as one of the product:



(C – 2 epimer of D – glucose)



C)



24. $\text{FeCr}_2\text{O}_4 + \text{Na}_2\text{CO}_3 + \text{O}_2 \xrightarrow{\text{Fusion}} [\text{X}] \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} [\text{Y}] \xrightarrow[\text{H}_2\text{O}_2 / \text{ether}]{\text{H}^+} [\text{Z}]$

Which of the following statements is true for the chromium compounds $[\text{X}]$, $[\text{Y}]$ and $[\text{Z}]$?

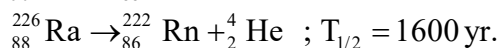
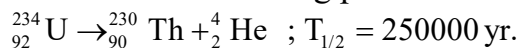
(I) In all three compounds, the chromium is in +6 oxidation state.

(II) $[\text{Z}]$ is a deep blue coloured compound which decomposes rapidly in aqueous solution into Cr^{3+} and dioxygen

(III) Saturated solution of $[\text{Y}]$ gives bright orange compound, chromic anhydride, with cold, concentrated H_2SO_4 .

- A) only I and II are correct B) only II and III are correct
 C) only III and I are correct D) All I, II, III are correct

25. Consider the following process of decays:



After above process has occurred for a long time, a state is reached where every two thorium atoms formed from ${}_{92}^{234}\text{U}$, one decomposes to form ${}_{88}^{226}\text{Ra}$ and for every two ${}_{88}^{226}\text{Ra}$ formed, one decomposes. Calculate the ratio of number of nuclei of ${}_{90}^{230}\text{Th}$ to ${}_{88}^{226}\text{Ra}$ at this state:

- A) 50 B) 100 C) 200 D) 25

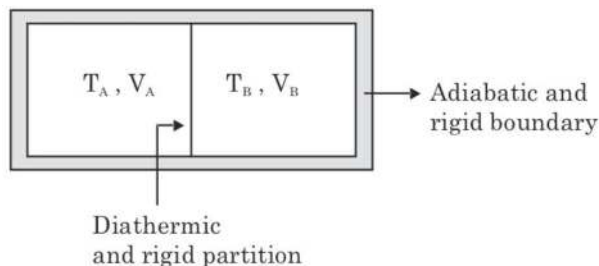
SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

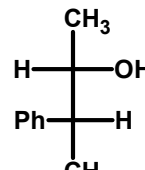
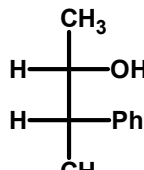
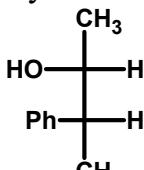
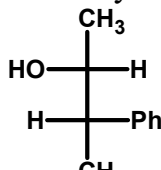
26.



Both compartments in the above diagram are large and contain ideal gas. The initial conditions are mentioned in the diagram. Select the correct statement(s) for the whole system.

- A) $\Delta S > 0$ if $T_A > T_B$
 B) $\Delta S > 0$ if $T_A < T_B$
 C) $\Delta S = 0$ if $T_A = T_B$
 D) $\Delta U = 0$ no matter that what is the relationship between T_A and T_B

27. The stereoisomer obtained in the reaction of (s)-2-phenyl propanal with MeMgBr followed by acid hydrolysis is/are



28. Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the following statements is/are correct for this entire process?

- A) Evolved I_2 is reduced B) CuI_2 is formed
 C) $\text{Na}_2\text{S}_2\text{O}_3$ is oxidised D) Cu_2I_2 is formed

29. Decomposition of A(g) follows order kinetics according to the reaction:

$\text{A(g)} \rightarrow 2\text{B(g)} + \text{C(g)}$ If initial pressure of A(g) is 10 atm and total pressure after 70 sec is 20 atm, then : ($\ln 2 = 0.7$)

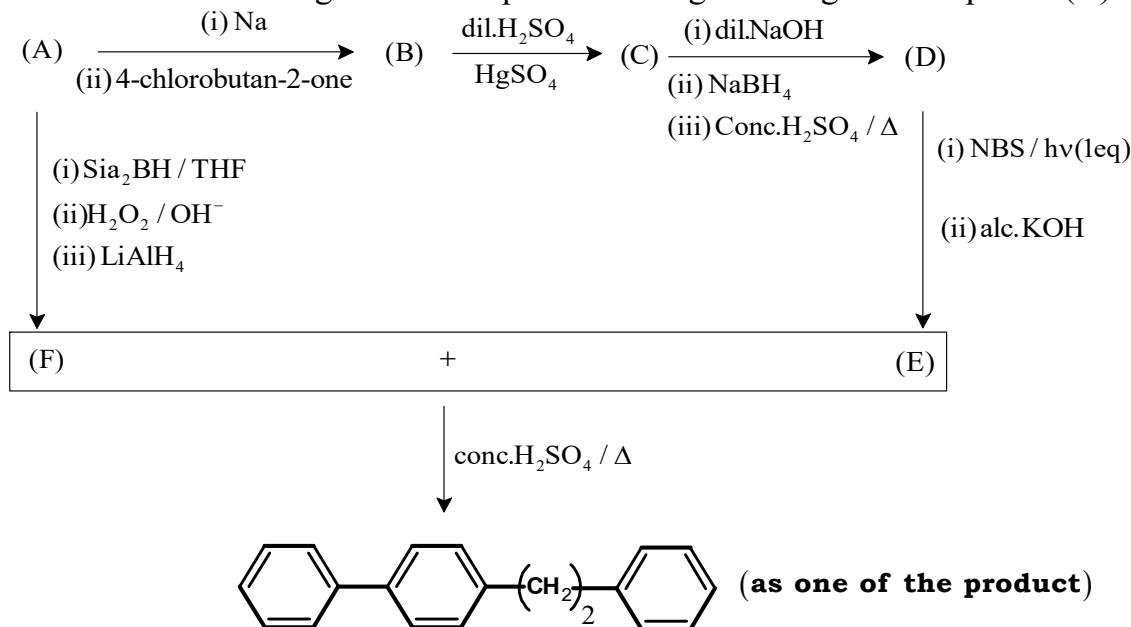
- A) Rate constant of reaction is 0.01 sec^{-1}
 B) Partial pressure of B at 70 sec is 10 atm
 C) Partial pressure of C at 140 sec is 7.5 atm.
 D) Initial rate of reaction is 0.1 atm/sec
30. Polymers formed by isoprene monomer on ozonolysis will give which of the following compounds as one of the possible product...
- A) $\text{HC}=\text{CH}_2-\text{CH}_2-\text{C}=\text{CH}_3$
 $\text{O} \quad \text{O}$
- B) $\text{H}_3\text{C}-\text{C}=\text{CH}_2-\text{CH}_2-\text{C}=\text{CH}_3$
 $\text{O} \quad \text{O}$
- C) $\text{HC}=\text{CH}_2-\text{CH}_2-\text{CH}=\text{O}$
 $\text{O} \quad \text{O}$
- D) $\text{H}_3\text{C}-\text{C}=\text{CH}_3$
 O
31. in the qualitative analysis of basic radicals, $\text{K}_4[\text{Fe}(\text{CN})_6]$ is used for the detection of
 A) Fe^{3+} B) Zn^{2+} C) Cu^{2+} D) Al^{3+}
32. Which of the following is/are correct for metallic crystal?
 A) Number of next nearest neighbour in FCC unit cell is 12.
 B) Packing efficiency of BCC unit cell is 68%
 C) Total number of TV and OV in one FCC unit cell is 8 and 4 respectively.
 D) Volume of HCP unit cell is $24\sqrt{2}R^3$, where R is radius of atom

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

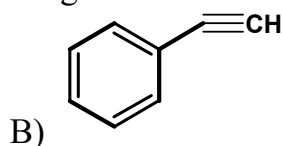
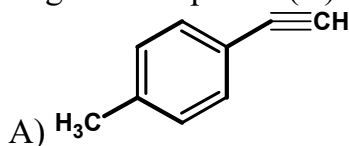
Paragraph For Questions 33 and 34

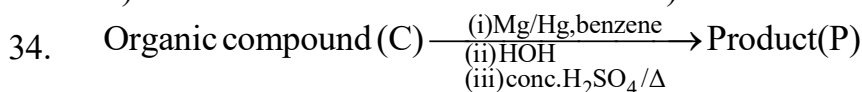
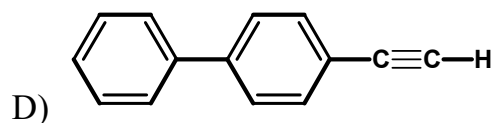
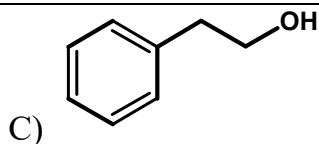
Consider the following reaction sequence starting from organic compound (A)



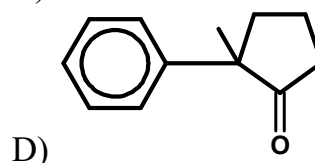
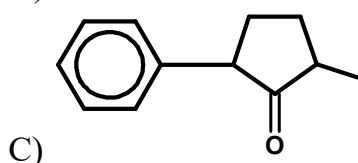
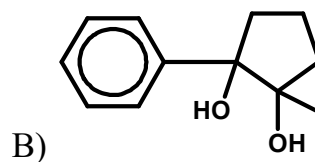
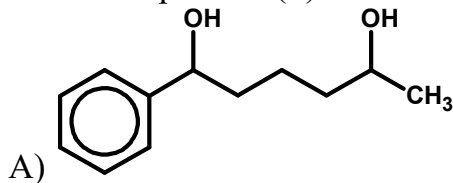
Now, answer the following questions:

33. Organic compound (A) in the above passage is:





Hence the product (P) is:



Paragraph For Questions 35 and 36

Weakest Lewis acid 'X' out of Boron halides (BX_3) when treated with metal hydride (Y) forms mixture of compounds. One of the compounds in this mixture when heated with ammonia forms another compound 'Z' which is aromatic in nature.

35. Reaction between 'X' and 'Y' produces mixture of compounds. Choose the correct properties of these possible compounds

- (I) one of the possible compounds is B_2H_6 which has maximum 6 atoms in one plane
- (II) one of the possible compounds is LiF which has lowest water solubility among fluorides of alkali metals
- (III) one of the possible compounds is AlF_3 which can be obtained by treating alumina with fluoro silicic acid

- A) I and II only
- B) II and III only
- C) I and III only
- D) all I, II and III

36. How many of the following statements are true about compounds in the passage

- I. When BeCl_2 is treated with Y it forms BeH_2
- II. All nitrogen and Boron atoms in molecule Z are in sp^2 hybridization
- III. Compound X has less than expected 'B-X' bond length

- A) Only I and II
- B) Only II and III
- C) Only II
- D) All I, II, and III

MATHS

SECTION - I

(SINGLE CORRECT CHOICE TYPE)

This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct.**

Marking scheme +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. The number of solution in x of the equation

$$\sqrt{\sqrt{3}f(x) - f'(x) - 2} + \sqrt{\cot 3x + \cos^2 x - 4} = \sin\left(\frac{3x}{2}\right) + \frac{\sqrt{2}}{2} \text{ if } x \in [0, 6\pi] \quad \text{and}$$

$f(x)$ satisfies the differential equation $f''(x) + f(x) = 0$ and

$$(f'(x))^2 + (f(x))^2 = 1, \text{ and } f(0) = 1 \text{ is}$$

- A) 1 B) 3 C) 4 D) 0

38. In a matrix $A_{4 \times 4}$, an ant starts moving from a_{11} in such a way that it can move from a_{ij} to either $a_{(i+1)j}$ or $a_{i(j+1)}$ in one step. If it reaches a_{44} , the probability that it passes through a_{23} is

- A) $\frac{18}{35}$ B) $\frac{1}{4}$ C) $\frac{9}{20}$ D) None of these

39. Three circle C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 2$ and $r_3 = 8$ they are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exist two straight lines each of which is a direct common tangent simultaneously to all the three circles then r_2 is equal to

- A) $r_2 = 4$ B) $r_2 = 5$ C) $r_2 = 10$ D) $r_2 = 16$

40. A plane passing through $(1, 1, 1)$ and parallel to the lines L_1, L_2 direction ratios $(1, 0, -1), (1, -1, 0)$. If V is the volume of tetrahedron formed by origin and the points where the plane intersect the coordinate axes, then value of $2V$ is equal to

- A) 8 B) 9 C) 10 D) 2

41. If a tangent of slope 4 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle

$$x^2 + y^2 + 4x + 1 = 0 \text{ then the maximum value of } ab \text{ is}$$

- A) 4 B) 5 C) 6 D) 8

42. If vectors $\hat{a}, \hat{b}, \hat{c}$ are unit vectors form RHS in space such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$ then $(\hat{a} \times \hat{b}) \cdot (\hat{a} \times (\hat{b} \times \hat{c}))$ is equal to

- A) $\frac{1}{\sqrt{2}}$ B) $-\frac{1}{\sqrt{2}}$ C) $\frac{1}{2\sqrt{2}}$ D) $-\frac{1}{2\sqrt{2}}$

43. Let A, B, C and D be four distinct point on a line in that order. The circle with diameters AC is $x^2 + y^2 + ax + c = 0$ and BD is $x^2 + y^2 - by = 0$ intersect at X and Y,

the line XY meets BC at Z. let P be a point on XY other than Z, the line CP intersects the circle with diameter AC at C and M, and line BP intersects the circle with diameter BD at B and N and the equation of the line AM and DN are $bx + cy + a = 0$ and $cx + ay + b = 0$ respectively, then which of the following is true? (Where ω is the complex cube root of unity)

A) $a + b + c = 1$

B) $a\omega + b\omega^2 + c = 1$

C) $a + b\omega^2 + c\omega = 0$

D) None of these

SECTION-II**(ONE OR MORE OPTIONS CORRECT TYPE)**

This section contains 7 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

44. Let $f(x) = 2x^3 - 3x^2 - x + \frac{3}{2}$ then which of the following is/are incorrect?
- A) $\int_{1/8}^{7/8} f(f(x)) dx = \frac{3}{4}$
- B) $y = (f(x))^{3/5}$ is not differentiable at exactly 1 point
- C) $y = [f(x) + \lambda]$, ($\lambda \in$ prime number, $[.] \rightarrow \text{GIF}$) is discontinuous at 3 points if $x \in [0, 1]$
- D) $f(f(x)) = 0$ has minimum 6 real roots
45. Which of the following is/are incorrect?
- A) Let $f: R \rightarrow R$, such that $f(x) = 2x + \left[\frac{x(x^2 - 1)}{4(x^4 - x^2 + 1)} + \frac{1}{8} \right] + [x] + \sin x \cos x$ then (where $[.]$ denotes the greatest integer function) f is one – one & onto:
- B) Let $f: R \rightarrow R$, such that $f(x) = \frac{x^3 + 2x^2}{\sin x + 2}$ then f is one – one & onto
- C) Let $f: R \rightarrow [1, \infty)$ such that $f(x) = 2^{|x-1|^{\sqrt{2}}}$ then $f(x)$ is one – one & into
- D) Let $f: R - \{0\} \rightarrow R$ such that $f(x) = |x| \ln |x|$ then $f(x)$ is one – one & onto
46. For any two positive integers x and y
- $$f(x, y) = \frac{1}{(x+1)!} + \frac{1}{(x+2)!} + \frac{1}{(x+3)!} + \dots + \frac{1}{(x+y)!},$$
- then which of the following options is/are correct
- A) $f(x, y) \leq \frac{1}{x} \left(\frac{1}{x!} - \frac{1}{(x+y)!} \right)$
- B) $\lim_{y \rightarrow \infty} (f(x, y)) \leq \frac{1}{x!}$
- C) $f(x, x) < \frac{1}{(x-1)!}$
- D) $f(2, 2)$ is equal to $\frac{2}{3}$
47. If largest and smallest value of $\frac{y-4}{x-3}$ are p and q respectively where (x, y) satisfy $x^2 + y^2 - 2x - 6y + 9 = 0$, then which of the following is/are true?

A) $p + q = \frac{4}{3}$ B) $q = 1$ C) $p = \frac{4}{3}$ D) $pq = \frac{4}{3}$

48. Consider a function

$$f(x) = \sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-x^2}{1+x^2} + \tan^{-1} \frac{2x}{1-x^2} - a \tan^{-1} x \quad (a \in R), \text{ the value of } a \text{ if}$$

$f(x) = 0$ for all x :

A) 6 B) -6 C) 2 D) -2

49. $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{3^m (n \cdot 3^m + m \cdot 3^n)} = \frac{p}{q}$ (where p & q are coprime) then which of the following

is/are correct?

A) $p + q = 41$ B) $p < q$
 C) p is perfect square D) q is perfect square

50. Let $A(4, 3)$, $B(-4, 3)$ and $C(0, -5)$ be the vertices of triangle and $P(5, 0)$ be a point. Let L , M and N be the feet of the perpendiculars drawn from P onto the sides BC , CA and AB respectively, then

- A) Area of $\triangle LMN$ is 5sq. units
 B) the centroid of $\triangle LMN$ does not exist
 C) The orthocenter of $\triangle LMN$ is the origin
 D) L , M and N are collinear

SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 51 and 52

Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of the equation

$$f(x) = x^n + p_1 x^{n-1} + \dots + p_{n-1} x + p_n = 0$$

$$\frac{f(x)}{x - \alpha_i} = x^{n-1} + (a_i + p_1)x^{n-2} + (a_i^2 + a_i p_1 + p_2)x^{n-3} + \dots \text{ then } \mathop{\text{a}}\limits_{i=1}^n \frac{f(x)}{x - \alpha_i} = f'(x)$$

is an identify where $f'(x) = \frac{df(x)}{dx}$

51. If $f(x) = x^5 + px^3 + qx + t = 0$, $\mathop{\text{a}}\limits_{i=1}^5 a_i^3$

A) $-p^3 + 3q$ B) 1 C) 0 D) $-p^3 + 7q$

52. If $f(x) = x^3 - 2x^2 + x - 1$ and α_i ($i = 1, 2, 3$), are three roots of $f(x) = 0$ then $\sum \alpha_i^3$ is equal to

A) 3 B) 4 C) 5 D) 6

Paragraph For Questions 53 and 54

A variable line L intersects the parabola $y = x^2$ at points P and Q whose x – coordinates are α and β respectively with $\alpha < \beta$, the area of the region enclosed by

the segment PQ and the parabola is always equal to $\frac{4}{3}$. The variable segment PQ has its middle point as M.

53. Which of the following is/are correct?

A) $(\beta - \alpha)$ can have more than one real values

B) $(\beta - \alpha)$ can be equal to 2

C) $(\beta - \alpha)$ can have exactly one real value

D) $\alpha = 2 + \beta$

54. Which of the following is/are correct?

A) Equations of the pair of tangents, drawn to the curve, represented by locus of M from origin are $y = 2x$ and $y = -2x$

B) Area of pair of tangents to be curve, represented by locus of M from origin are $y = x$ and $y = -x$

C) Area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it from origin is $\frac{2}{3}$ sq. units

D) Area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it from origin is $\frac{1}{3}$ sq. units

1	C	2	A	3	C	4	C	5	A
6	D	7	C	8	ACD	9	ABC	10	ABC
11	ACD	12	CD	13	ABD	14	AB	15	B
16	A	17	C	18	C				
19	A	20	B	21	B	22	D	23	A
24	D	25	B	26	ABCD	27	AC	28	ACD
29	ABCD	30	ABC	31	ABC	32	BCD	33	B
34	D	35	D	36	D				
37	A	38	C	39	A	40	B	41	D
42	D	43	C	44	ABC	45	ABCD	46	ABC
47	AC	48	ACD	49	ABC	50	BD	51	C
52	C	53	B	54	C				